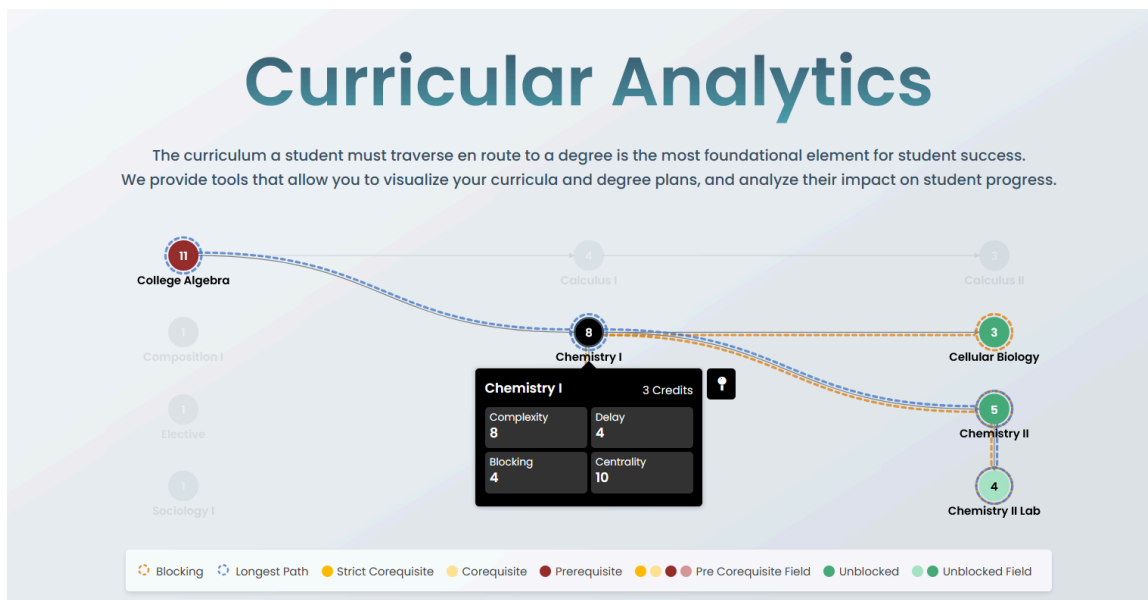


What is Curricular Analytics

An open-source framework and methodology designed to visualize, quantify, and analyze the structural complexity of academic programs.

Instead of just looking at a degree as a checklist of classes, it treats a curriculum as a network (a directed acyclic graph) to understand how course requirements impact student progression and graduation rates.



How does it normally work: Upload

Uploading Curricula and Degree Plans - File Formats

In order to upload a curriculum or degree plan to the Curricular Analytics site, you simply need to specify them in a CSV file, and then upload them to your account as described above.

The screenshot shows a Google Sheet titled "curriculum-ex" with a menu bar (Home, Insert, Page Layout, Formulas, Data, Review, View) and a search bar. The sheet contains a curriculum plan for "Underwater Basket Weaving" at "ACME State University". The plan includes a list of courses with their IDs, names, prefixes, numbers, prerequisites, corequisites, strict-corequisites, credit hours, institution, and canonical names.

Course ID	Course Name	Prefix	Number	Prerequisites	Corequisites	Strict-Corequisites	Credit Hours	Institution	Canonical Name
1	Introduction to Baskets	BW	101				3		
2	Swimming	PE	115				3		
3	Introductory Calculus w/ Basketry A	MA	116				4		
4	Basic Basket Forms	BW	111	1			3		
5	Basic Basket Forms Lab	BW	111L			5	1		
6	Advanced Basketry	BW	201	4;5	3		3		
7	Basket Materials & Decoration	BW	214	1			3		
8	Underwater Weaving	BW	301	2	7		3		

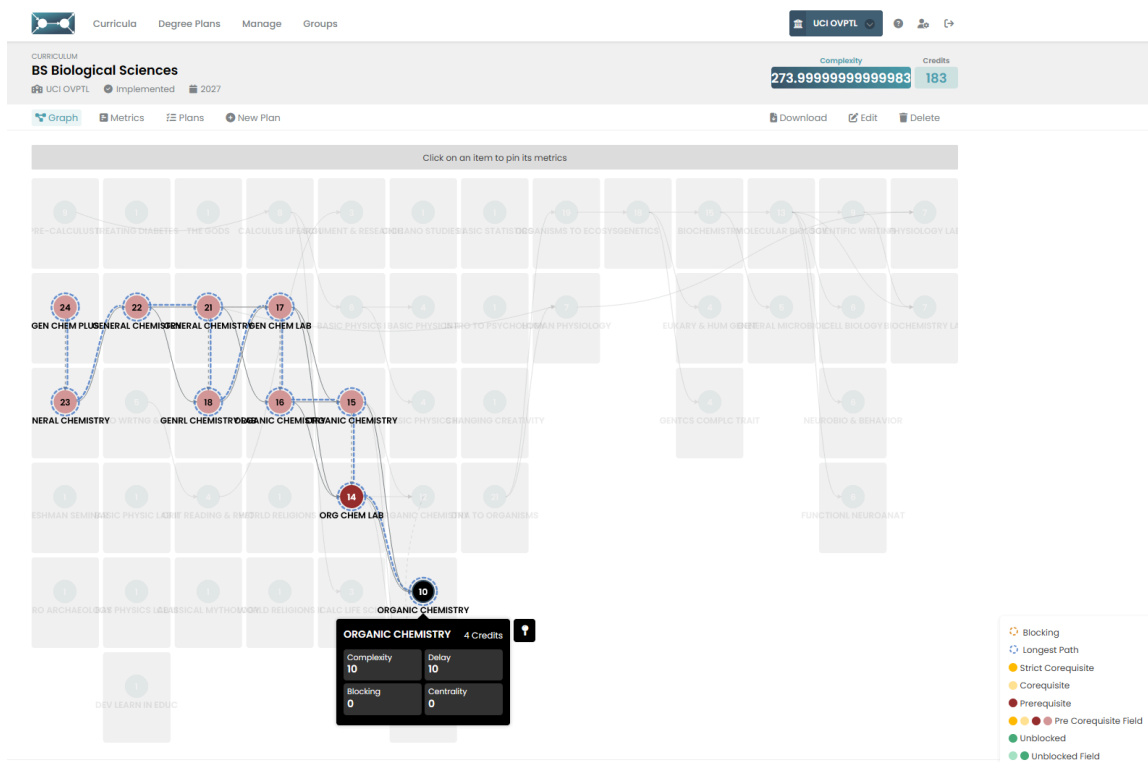
The screenshot shows a Google Sheet titled "degree-plan-ex" with a menu bar (Home, Insert, Draw, Page Layout, Formulas, Data, Review, Tell me) and a search bar. The sheet contains a degree plan for "Underwater Basket Weaving" at "ACME State University". The plan includes a list of courses with their IDs, names, prefixes, numbers, prerequisites, corequisites, strict-corequisites, credit hours, institution, canonical names, and terms.

Course ID	Course Name	Prefix	Number	Prerequisites	Corequisites	Strict-Corequisites	Credit Hours	Institution	Canonical Name	Term
1	Introduction to Baskets	BW	101				3			1
2	Swimming	PE	115				3			1
3	Introductory Calculus w/ Basketry A	MA	116	9			4			2
4	Basic Basket Forms	BW	111	1			3			2
5	Basic Basket Forms Lab	BW	111L			5	1			2
6	Advanced Basketry	BW	201	4;5	3		3			3
7	Basket Materials & Deco	BW	214	1			3			3
8	Underwater Weaving	BW	301	2	7		3			3

1. Definition & Scope

- ## 2. Relationship (One-to-Many)

- ## Output

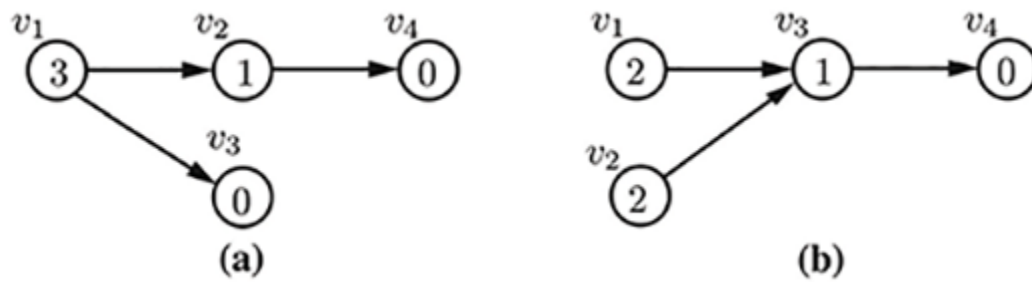


Curricula Metrics

Blocking Factor

The blocking factor is an important curriculum-based metric because it measures the extent to which one course blocks the ability to take other courses in the curriculum. That is, a course with a high blocking factor acts as a gateway to many other courses in the curriculum. Students who are unable to pass the gateway course will be blocked from taking many other courses in the curriculum.

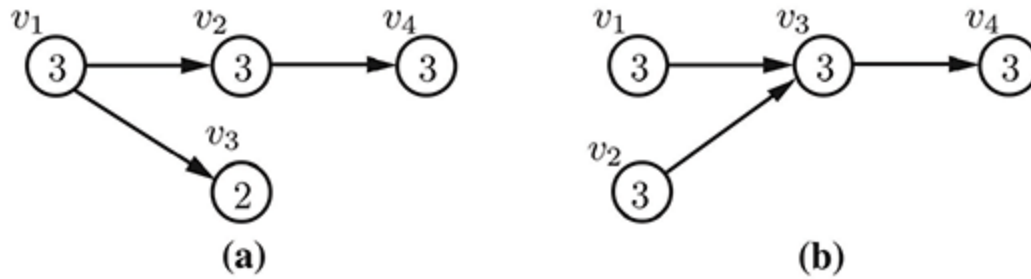
The blocking factor of each course is shown inside of the course vertices in this figure.



Delay Factor

Many curricula, particularly those in science, technology engineering and math (STEM) fields, contain a set of courses that must be completed in sequential order. The ability to successfully navigate these long pathways without delay is critical for student success and on-time graduation. If any course on the pathway is not completed on time, the student will then be delayed in completing the entire pathway by one term. The delay factor metric allows us to quantify this effect.

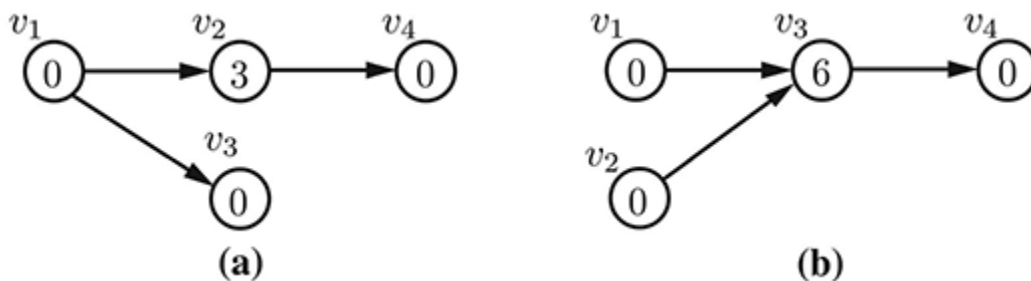
. As an example of the delay factor metric, consider the same four-course curricula shown above. The delay factor of each course is shown inside of the course vertices in the figure below.



Centrality

A course can be thought of as being central to a curriculum if it requires a number of foundational courses as prerequisites, and the course itself serves as a prerequisite to many additional discipline-specific courses in the curriculum. The centrality metric is meant to capture this notion.

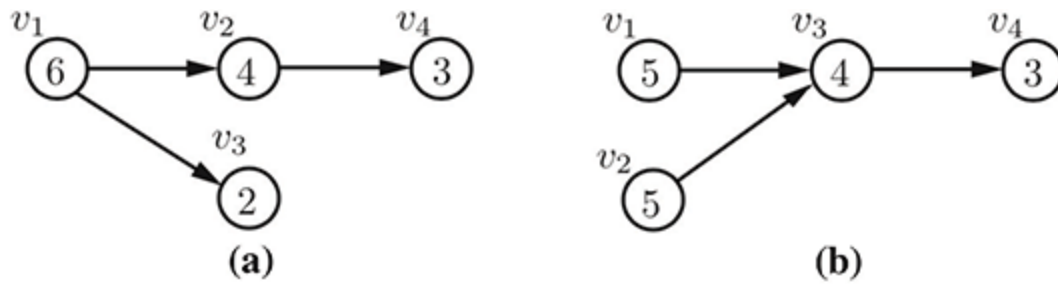
We define the centrality of source and sink vertices to be 0. For all other course vertices, consider all of the long paths (i.e., unique paths from a source to a sink) containing course vertex v is given by the sum of these path lengths. As an example of the centrality metric, consider the same four-course curricula shown above. The centrality factor of each course is shown inside of the vertices in the figure below. In the case of the curriculum in part (a), there is one long path of length three that includes course v_2 hence its centrality is 3, while in part (b), there are two long paths of length three that include course v_3 , hence its centrality is 6.



Structural Complexity

The curricular complexity of a course is meant to capture the impact of curricular structure on student progression. Through experimentation, we have found that a simple linear combination of the delay and blocking factors described above provides a good measure for quantifying the structural complexity of a curriculum. Specifically, we have found a high correlation between increased structural complexity and decreased graduation rates.

As an example of the structural complexity metric, consider the same four-course curricula shown above. The complexity factor of each course, which is simply the sum of the course's delay and blocking factors, are shown inside of the course vertices in this figure.



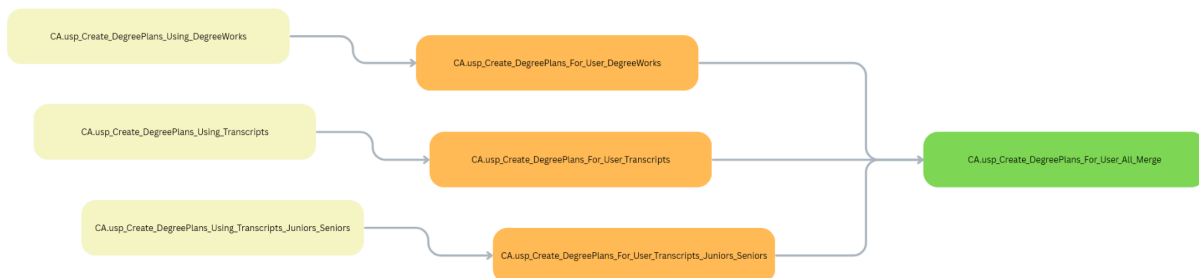
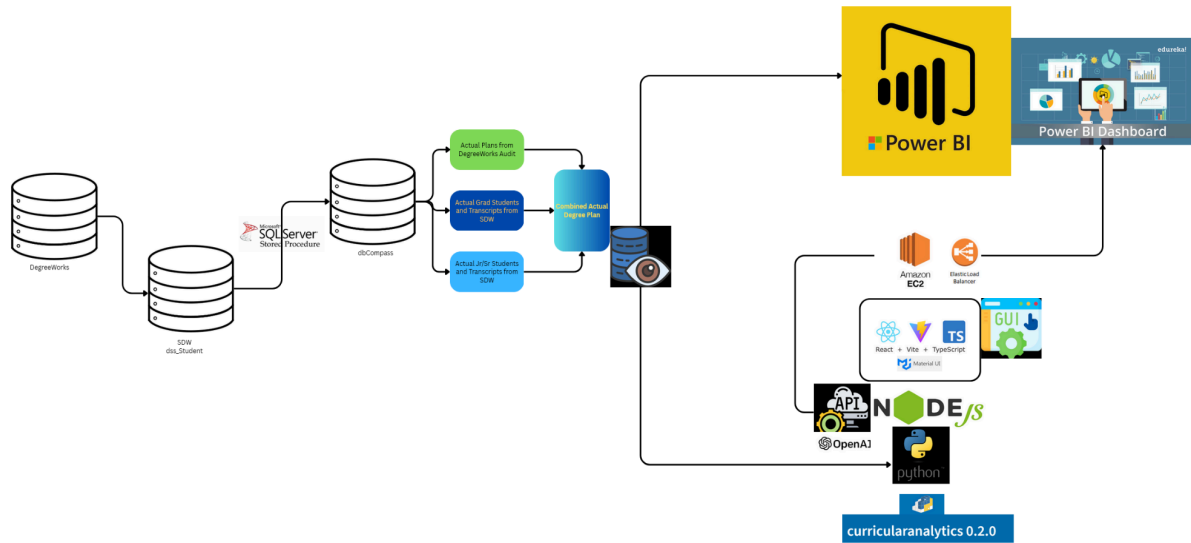
APIs Available

- ☐ [CurricularAnalytics library for R](#)
- ☐ [CurricularAnalytics for Julia NoteBooks](#)
- ☒ [Python Port from the Julia Version - Maintained by UCSD](#)

Main [API Methods](#) used

- Reading/writing curricula and degree plans: [write_csv\(\)](#), [read_csv\(\)](#)
- Metrics:
 - [blocking_factor\(\)](#)
 - [delay_factor\(\)](#)
 - [centrality\(\)](#)
 - [complexity\(\)](#)

How we process data



Available Data (dbCompass in sdw-d-db.aws.uci.edu)

[dbcompass].[CA].[TB_Actual_DegreePlan_ANON_CA_Metrics]

SQL

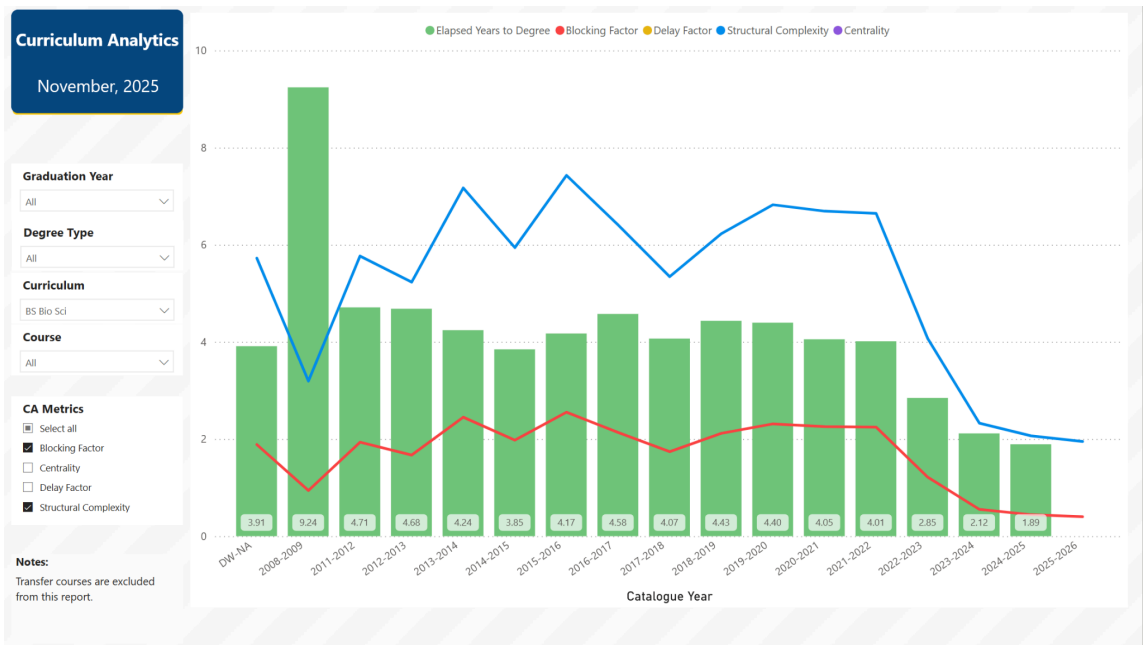
```
SELECT TOP (1000) [Student_ID_ANON]
, [Institution]
, [AcademicYear]
, [Curriculum]
, [DegreePlan]
, [DegreeType]
, [SystemType]
, [CIP]
, [CourseSequenceID]
, [CourseName]
, [Prefix]
, [Number]
, [Prerequisites]
, [Corequisites]
, [Strict_Corequisites]
, [CreditHours]
, [CanonicalName]
, [TermNumber]
, [Course_URL]
, [Plan_URL]
, [Notes]
, [ExtractDate]
, [DegreePlanID]
, [CA_Blocking_Factor]
, [CA_Delay_Factor]
, [CA_Centrality]
, [CA_Structural_Complexity]
, [CA_MetricDate]
, [DB_INSERT_TIMESTAMP]
FROM [dbcompass].[CA].[TB_Actual_DegreePlan_ANON_CA_Metrics]
WHERE Student_ID_ANON='S00530182'
order by [CourseSequenceID]
```


CourseSequenceID	CourseName	Prefix	Number	Prerequisites	Corequisites	Strict_Corequisites	CreditsHours	CanonicalName	TermNumber	Course_URL	Plan_URL	Notes	ExtractDate	DegreePlanID	CA_Blocking_Factor	CA_Delay_Factor	CA_Centrality	CA_Structural_Complexity
6	PSYCH 10A	PSYCH	10A				4	PSYCH10A	1			Prereqs and Coreqs completed	2008-03-01 21:25:57 023	9	2	2	0	4
7	PSYCH 9A	PSYCH	9A				4	PSYCH9A	1			Prereqs and Coreqs completed	2008-03-01 21:25:57 023	9	3	2	0	5
8	SOCOL 1	SOCOL	1				4	SOCOL1	1			Prereqs and Coreqs completed	2008-03-01 21:25:57 023	9	0	1	0	1
9	HISTORY 36C	HISTORY	36C				4	HISTORY36C	2			Prereqs and Coreqs completed	2008-03-01 21:25:57 023	9	0	1	0	1
10	PSYCH 10B	PSYCH	10B				4	PSYCH10B	2			Prereqs and Coreqs completed	2008-03-01 21:25:57 023	9	0	2	0	2
11	PSYCH 9C	PSYCH	9C				4	PSYCH9C	2			Prereqs and Coreqs completed	2008-03-01 21:25:57 023	9	1	2	0	3
12	WRITING 45	WRITING	45				6	WRITING45	2			Prereqs and Coreqs completed	2008-03-01 21:25:57 023	9	1	2	0	3
13	PSYCH 10C	PSYCH	10C				4	PSYCH10C	3			Prereqs and Coreqs completed	2008-03-01 21:25:57 023	9	0	2	0	2
14	PSYCH 14P	PSYCH	14P				4	PSYCH14P	3			Prereqs and Coreqs completed	2008-03-01 21:25:57 023	9	0	1	0	1
15	PSYCH 9B	PSYCH	9B				4	PSYCH9B	3			Prereqs and Coreqs completed	2008-03-01 21:25:57 023	9	4	2	0	6
16	WRITING 60	WRITING	60				4	WRITING60	3			Prereqs and Coreqs completed	2008-03-01 21:25:57 023	9	0	2	0	2
17	CRM.LAW C7	CRM.LAW	C7				4	CRM.LAWC7	4			Prereqs and Coreqs completed	2008-03-01 21:25:57 023	9	0	1	0	1
18	MATH 1A	MATH	1A				0	MATH1A	4			Prereqs and Coreqs completed	2008-03-01 21:25:57 023	9	2	3	0	5
19	COM LIT 60A	COM LIT	60A				4	COM LIT60A	5			Prereqs and Coreqs completed	2008-03-01 21:25:57 023	9	0	1	0	1
20	BIO SCI 36	BIO SCI	36				4	BIO SCI36	6			Prereqs and Coreqs completed	2008-03-01 21:25:57 023	9	0	1	0	1
21	CRM.LAW C10	CRM.LAW	C10				4	CRM.LAWC10	6			Prereqs and Coreqs completed	2008-03-01 21:25:57 023	9	0	1	0	1
22	MATH 1B	MATH	1B				4	MATH1B	6			Prereqs and Coreqs completed	2008-03-01 21:25:57 023	9	1	3	3	4
23	PSYCH 120A	PSYCH	120A				4	PSYCH120A	6			Prereqs and Coreqs completed	2008-03-01 21:25:57 023	9	0	2	0	2
24	BIO SCI 41	BIO SCI	41				4	BIO SCI41	7			Prereqs and Coreqs completed	2008-03-01 21:25:57 023	9	0	1	0	1
25	CLASSIC 45B	CLASSIC	45B				4	CLASSIC45B	7			Prereqs and Coreqs completed	2008-03-01 21:25:57 023	9	0	1	0	1
26	MATH 2A	MATH	2A				4	MATH2A	7			Prereqs and Coreqs completed	2008-03-01 21:25:57 023	9	0	3	0	3
27	PSYCH 160A	PSYCH	160A				4	PSYCH160A	7			Prereqs and Coreqs completed	2008-03-01 21:25:57 023	9	0	2	0	2
28	BIO SCI 37	BIO SCI	37				4	BIO SCI37	8			Prereqs and Coreqs completed	2008-03-01 21:25:57 023	9	0	1	0	1
29	LPS 51	LPS	51				4	LPS51	8			Prereqs and Coreqs completed	2008-03-01 21:25:57 023	9	0	1	0	1
30	PSYCH 140M	PSYCH	140M				4	PSYCH140M	8			Prereqs and Coreqs completed	2008-03-01 21:25:57 023	9	0	2	0	2
31	PSYCH 160D	PSYCH	160D				4	PSYCH160D	8			Prereqs and Coreqs completed	2008-03-01 21:25:57 023	9	0	2	0	2
32	COGS 140J	COGS	140J				4	COGS140J	9			Prereqs and Coreqs completed	2008-03-01 21:25:57 023	9	0	2	0	2
33	COGS 177D	COGS	177D				4	COGS177D	9			Prereqs and Coreqs completed	2008-03-01 21:25:57 023	9	0	1	0	1
34	CRM.LAW C104	CRM.LAW	C104				4	CRM.LAWC104	9			Prereqs and Coreqs completed	2008-03-01 21:25:57 023	9	0	1	0	1

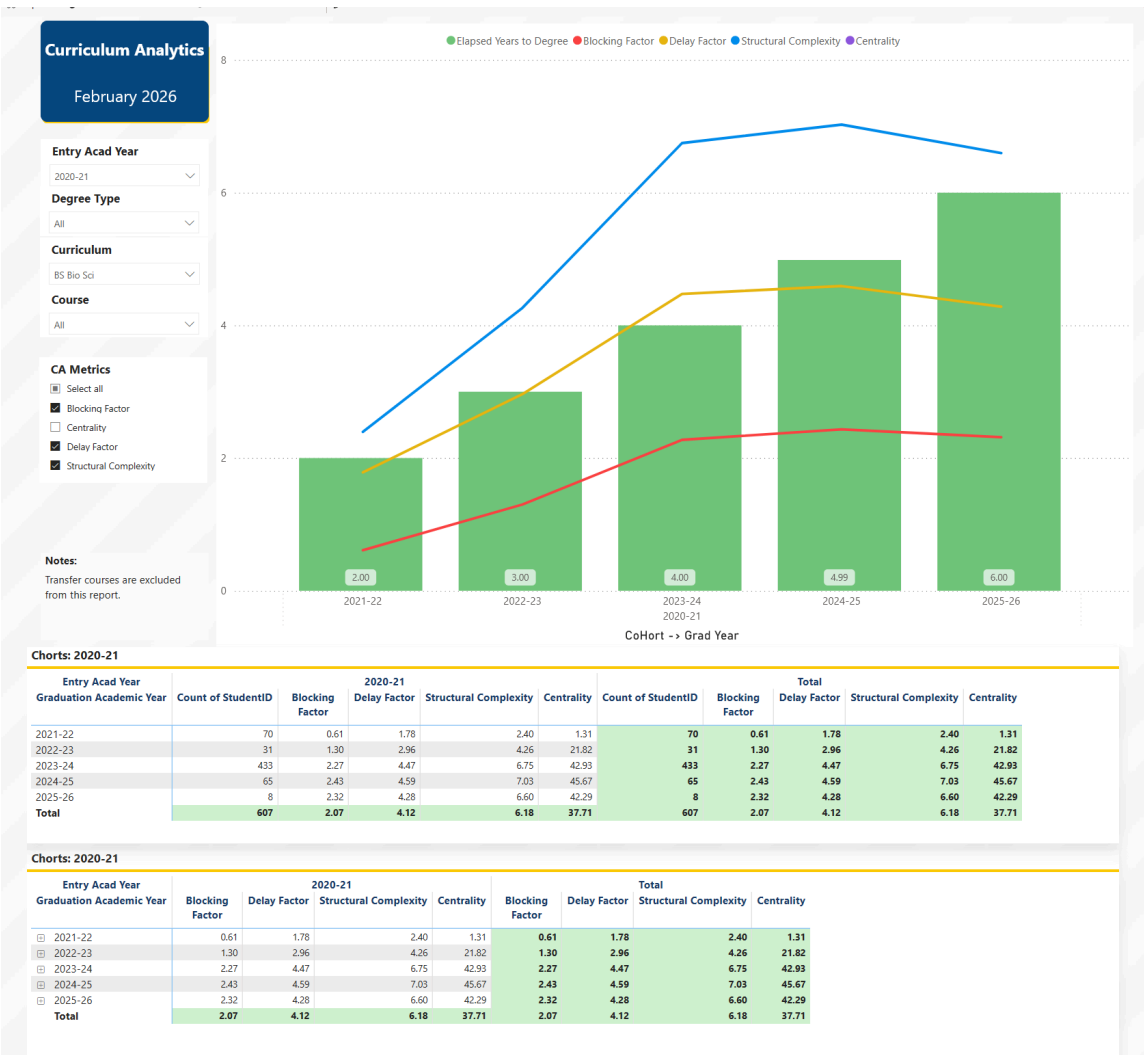
How data can be used

PowerBI

Analysis by Catalog Year



Analysis by Entry Year



Curriculum/Degree Plan

Curriculum Analytics

February 2026

Catalogue Year

2019-2020

Degree Type

BS

Curriculum

All

Course

All

Students

12000

Student ID

Search

FAQ: What is Catalogue Year?

This report shows the Catalogue Year associated with the student's school/major requirements.

Courses with student count between 1 and 2,000

Catalogue Year	Curriculum	Course Num	Course	Credit Hours	Students	Blocking Factor	Centrality	Delay Factor	Co
2019-2020	BS Engr BM	MATH 1B	MATH 1B	4.00	2	40.50	170.50	10.00	
2019-2020	BS Engr CE	MATH 1B	MATH 1B	4.00	3	37.67	101.33	11.00	
2019-2020	BS Engr CE	MATH 2A	MATH 2A	4.00	4	37.00	229.00	10.75	
2019-2020	BS Physics	MATH 2A	MATH 2A	4.00	5	36.40	51.40	7.20	
2019-2020	BS Engr EE	MATH 2A	MATH 2A	3.43	6	35.71	260.00	11.29	
2019-2020	BS Engr EE	MATH 2B	MATH 2B	3.71	13	35.43	1,381.14	12.43	
2019-2020	BS EngrChm	MATH 2A	MATH 2A	4.00	3	35.33	0.00	11.67	
2019-2020	BS Engr AE	MATH 2A	MATH 2A	4.00	5	34.80	0.00	8.00	
2019-2020	BS Engr BM	MATH 2A	MATH 2A	3.64	10	34.18	78.36	9.09	
2019-2020	BS MSE	MATH 2B	MATH 2B	4.00	3	34.00	72.33	9.00	
2019-2020	BS Engr ME	MATH 2A	MATH 2A	4.00	16	32.56	22.31	7.50	
2019-2020	BS EngrCpE	MATH 2A	MATH 2A	3.60	9	32.30	236.10	10.40	
2019-2020	BS EngrEnv	MATH 1B	MATH 1B	4.00	2	32.00	0.00	9.00	
2019-2020	BS Engr AE	MATH 2B	MATH 2B	4.00	7	32.00	266.00	7.71	
2019-2020	BS Physics	MATH 2B	MATH 2B	3.27	9	31.82	145.45	6.64	
2019-2020	BS EngrEnv	MATH 2A	MATH 2A	4.00	2	31.00	196.00	9.00	
2019-2020	BS EngrChm	MATH 2B	MATH 2B	4.00	4	31.00	841.25	11.25	
2019-2020	BS Engr ME	MATH 2B	MATH 2B	3.88	31	30.88	146.06	7.44	
2019-2020	BS Engr CE	MATH 2B	MATH 2B	3.60	9	30.00	117.20	9.00	
2019-2020	BS EngrEnv	MATH 2B	MATH 2B	4.00	2	29.00	193.00	9.00	
2019-2020	BS EngrCpE	MATH 1B	MATH 1B	4.00	4	28.25	0.00	12.00	
2019-2020	BS EngrChm	CHEM 1A	CHEM 1A	4.00	5	28.20	534.80	11.00	
2019-2020	BS Math	CHEM 2A	CHEM 2A	3.56	9	27.78	13.00	6.44	
2019-2020	BS Engr BM	MATH 2B	MATH 2B	3.75	15	27.75	169.19	8.81	
2019-2020	BS EngrCpE	EECS 12	EECS 12	4.00	17	27.00	60.47	8.00	
2019-2020	BS EngrChm	CHEM 1B	CHEM 1B	4.00	8	26.25	541.38	10.38	
2019-2020	BS Chem	MATH 2A	MATH 2A	3.33	10	25.83	314.75	10.08	
2019-2020	BS MSE	ENGR 1A	ENGR 1A	4.00	2	25.50	0.00	8.50	
2019-2020	BS EngrCpE	MATH 2B	MATH 2B	4.00	12	25.17	405.67	11.00	
2019-2020	BS CSE	I&C SCI 31	I&C SCI 31	4.00	4	25.00	0.00	7.75	
2019-2020	BS Engr EE	PHYSICS 7C	PHYSICS 7C	3.53	15	25.00	899.18	11.06	
Total				3.47	656	2.68	63.12	4.56	

>> Courses

Drill down to Course

2019-2020 BS Physics - MATH 2A

Catalogue Year	Curriculum	Course Num	Course	Student ID	Credit Hours	Blocking Factor	Centrality	Delay Factor
2019-2020	BS Physics	MATH 2A	MATH 2A	1012109	4.00	42.00	0.00	7.00
2019-2020	BS Physics	MATH 2A	MATH 2A	14431060	4.00	32.00	0.00	7.00
2019-2020	BS Physics	MATH 2A	MATH 2A	10131070	4.00	37.00	0.00	7.00
2019-2020	BS Physics	MATH 2A	MATH 2A	10140047	4.00	31.00	257.00	8.00
2019-2020	BS Physics	MATH 2A	MATH 2A	10150102	4.00	40.00	0.00	7.00
Total					4.00	36.40	51.40	7.20

Course Selected

>> Student

Drill down to Student

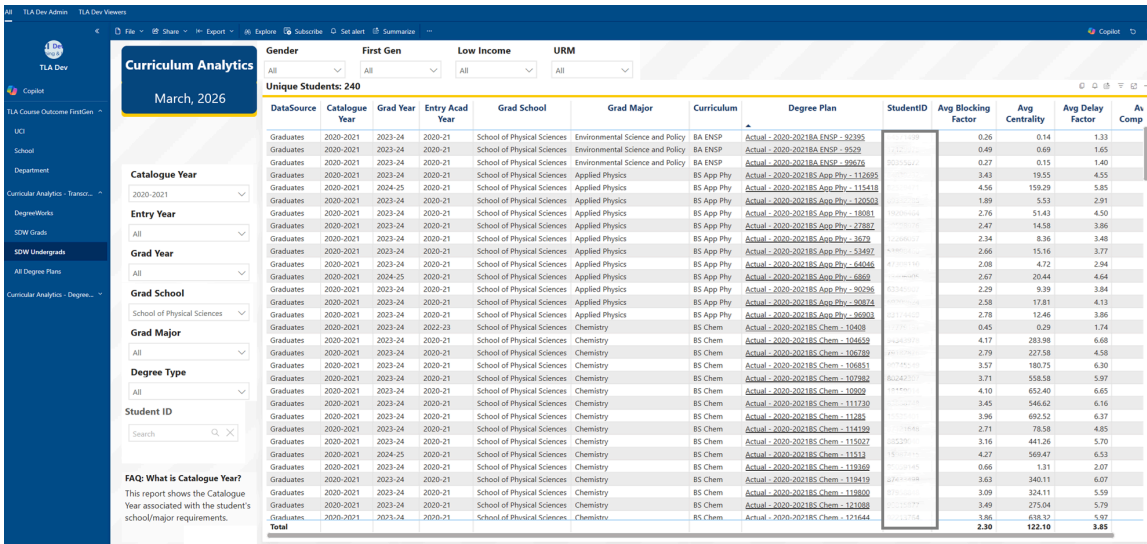
CA Metric for Student 24417898									
Catalogue Year	Curriculum	Student ID	Sequence	Course Num	Course Name	Credit Hours	Blocking Factor	Centrality	Delay Factor
2019-2020	BS Physics	24417898	3	AC ENG 22B	AC ENG 22B	2.00	0.00	0.00	1.00
2019-2020	BS Physics	24417898	4	UNI STU 84	UNI STU 84	4.00	0.00	0.00	1.00
2019-2020	BS Physics	24417898	5	AC ENG 20A	AC ENG 20A	5.00	2.00	0.00	3.00
2019-2020	BS Physics	24417898	6	AC ENG 22A	AC ENG 22A	2.00	0.00	0.00	1.00
2019-2020	BS Physics	24417898	7	MATH 2A	MATH 2A	4.00	32.00	0.00	7.00
2019-2020	BS Physics	24417898	8	PHYSICS 2	PHYSICS 2	4.00	0.00	0.00	2.00
2019-2020	BS Physics	24417898	9	STATS 7	STATS 7	4.00	0.00	0.00	1.00
2019-2020	BS Physics	24417898	10	AC ENG 20B	AC ENG 20B	5.00	1.00	3.00	3.00
2019-2020	BS Physics	24417898	11	MATH 2B	MATH 2B	4.00	30.00	304.00	7.00
2019-2020	BS Physics	24417898	12	PHYSICS 7C	PHYSICS 7C	4.00	22.00	115.00	7.00
2019-2020	BS Physics	24417898	13	PHYSICS 7LC	PHYSICS 7LC	1.00	0.00	0.00	4.00
2019-2020	BS Physics	24417898	14	AC ENG 20C	AC ENG 20C	0.00	0.00	0.00	3.00
2019-2020	BS Physics	24417898	15	I&C SCI 6N	I&C SCI 6N	4.00	0.00	0.00	1.00
2019-2020	BS Physics	24417898	16	POL SCI 21A	POL SCI 21A	4.00	0.00	0.00	1.00
2019-2020	BS Physics	24417898	17	PSYCH 9B	PSYCH 9B	4.00	0.00	0.00	1.00
2019-2020	BS Physics	24417898	18	STATS 7	STATS 7	0.00	0.00	0.00	1.00
2019-2020	BS Physics	24417898	19	ANTHRO 2A	ANTHRO 2A	4.00	0.00	0.00	1.00
2019-2020	BS Physics	24417898	20	MATH 2D	MATH 2D	4.00	1.00	4.00	4.00
2019-2020	BS Physics	24417898	21	PHYSICS 7E	PHYSICS 7E	4.00	16.00	161.00	7.00
2019-2020	BS Physics	24417898	22	WRITING 39A	WRITING 39A	4.00	2.00	0.00	3.00
2019-2020	BS Physics	24417898	23	MATH 2D	MATH 2D	0.00	21.00	88.00	7.00
2019-2020	BS Physics	24417898	24	MATH 2E	MATH 2E	4.00	0.00	0.00	4.00
2019-2020	BS Physics	24417898	25	PHYSICS 7D	PHYSICS 7D	4.00	7.00	68.00	7.00
2019-2020	BS Physics	24417898	26	PHYSICS 7LD	PHYSICS 7LD	1.00	0.00	0.00	5.00
2019-2020	BS Physics	24417898	27	WRITING 39B	WRITING 39B	4.00	1.00	3.00	3.00
2019-2020	BS Physics	24417898	28	HISTORY 21C	HISTORY 21C	4.00	0.00	0.00	1.00
2019-2020	BS Physics	24417898	29	MATH 3D	MATH 3D	4.00	3.00	11.00	6.00
2019-2020	BS Physics	24417898	30	WRITING 39C	WRITING 39C	4.00	0.00	0.00	3.00
2019-2020	BS Physics	24417898	31	GEN&SEX 50A	GEN&SEX 50A	4.00	0.00	0.00	1.00
Total						196.00	3.10	21.50	4.24

>> Transcripts

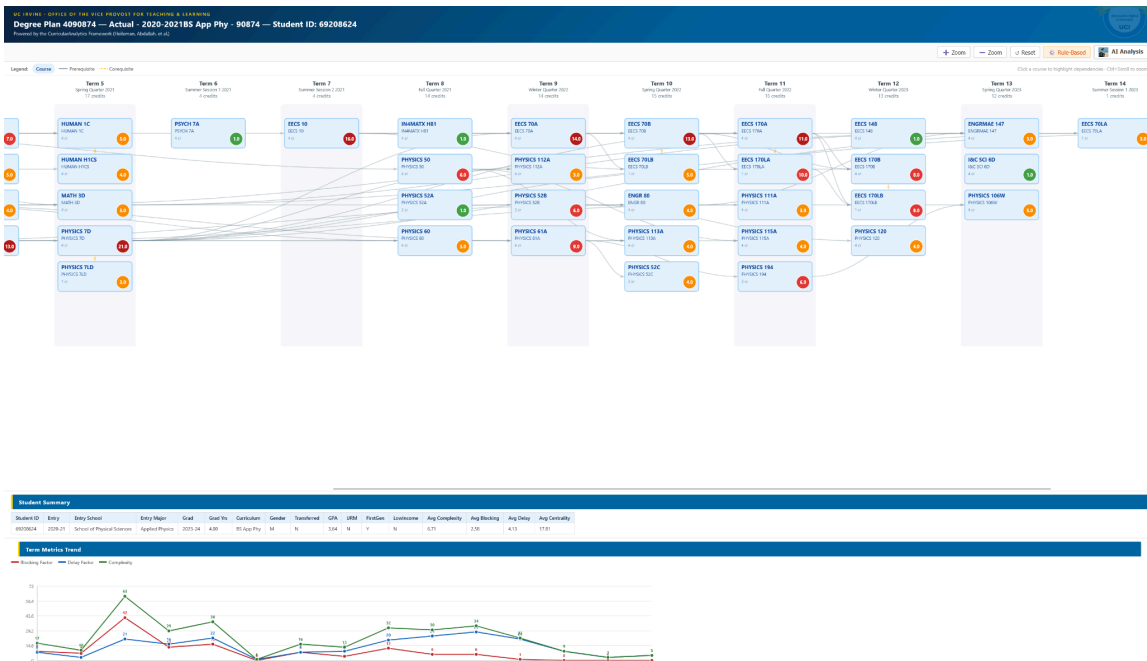
Drill down to Transcript

SDW Native Transcript for 24417898								
Student ID	Transcript Line Sort	Term	School	Course	Units Comp Graded	Grade	Grade Point	StudentID
24417898	711	Summer 2019	School of Humanities	AC ENG 22B	0.00	P	0.000	24417898
24417898	712	Summer 2019	Div of Undergraduate Education	UNI STU 84	4.00	A-	3.700	24417898
24417898	716	Fall Quarter 2019	School of Humanities	AC ENG 20A	0.00	P	0.000	24417898
24417898	717	Fall Quarter 2019	School of Humanities	AC ENG 22A	2.00	B+	3.300	24417898
24417898	718	Fall Quarter 2019	School of Info & Computer Sci	STATS 7	4.00	C-	0.000	24417898
24417898	719	Fall Quarter 2019	School of Physical Sciences	MATH 2A	4.00	C	2.000	24417898
24417898	720	Fall Quarter 2019	School of Physical Sciences	PHYSICS 2	4.00	A-	3.700	24417898
24417898	721	Winter Quarter 2020	School of Humanities	AC ENG 20B	0.00	P	0.000	24417898
24417898	722	Winter Quarter 2020	School of Physical Sciences	MATH 2B	4.00	B+	3.300	24417898
24417898	723	Winter Quarter 2020	School of Physical Sciences	PHYSICS 7C	4.00	B-	2.700	24417898
24417898	724	Winter Quarter 2020	School of Physical Sciences	PHYSICS 7LC	1.00	A	4.000	24417898
24417898	725	Winter Quarter 2020	Pharmacy & Pharmaceutical Sciences	PHRMSCI 42	1.00	A	4.000	24417898
24417898	726	Spring Quarter 2020	School of Humanities	AC ENG 20C	0.00	P	0.000	24417898
24417898	727	Spring Quarter 2020	School of Info & Computer Sci	I&C SCI 6N	4.00	A	4.000	24417898
24417898	728	Spring Quarter 2020	School of Social Sciences	POL SCI 21A	0.00	P	0.000	24417898
24417898	729	Spring Quarter 2020	School of Social Sciences	PSYCH 9B	4.00	A-	3.700	24417898
24417898	730	Summer 2020	School of Info & Computer Sci	STATS 7	0.00	A	4.000	24417898
24417898	731	Fall Quarter 2020	School of Humanities	WRITING 39A	0.00	P	0.000	24417898
24417898	732	Fall Quarter 2020	School of Physical Sciences	MATH 2D	4.00	D+	0.000	24417898
24417898	733	Fall Quarter 2020	School of Physical Sciences	PHYSICS 7E	4.00	A-	3.700	24417898
24417898	734	Fall Quarter 2020	School of Social Sciences	ANTHRO 2A	4.00	A	4.000	24417898
24417898	735	Winter Quarter 2021	School of Humanities	WRITING 39B	0.00	P	0.000	24417898
24417898	736	Winter Quarter 2021	School of Physical Sciences	MATH 2D	0.00	A-	3.700	24417898
24417898	737	Winter Quarter 2021	School of Physical Sciences	MATH 2E	4.00	A	4.000	24417898
24417898	738	Winter Quarter 2021	School of Physical Sciences	PHYSICS 7D	4.00	C+	2.300	24417898
24417898	739	Winter Quarter 2021	School of Physical Sciences	PHYSICS 7LD	1.00	A	4.000	24417898
24417898	740	Spring Quarter 2021	School of Humanities	HISTORY 21C	4.00	B+	3.300	24417898
24417898	741	Spring Quarter 2021	School of Humanities	WRITING 39C	0.00	P	0.000	24417898
24417898	742	Spring Quarter 2021	School of Physical Sciences	MATH 3D	4.00	A	4.000	24417898
24417898	743	Summer 2021	School of the Arts	MUSIC 3	4.00	A	4.000	24417898
24417898	744	Summer 2021	School of Humanities	GEN&SEX 50A	0.00	P	0.000	24417898
24417898	745	Fall Quarter 2021	School of Physical Sciences	PHYSICS 50	4.00	A	4.000	24417898
Total	746	Fall Quarter 2021	School of Physical Sciences	PHYSICS 50A	0.00	A-	3.000	24417898
					161.00		2.315	

PowerBI App



Drill thru CA Metrics



URL

<https://ca-dev.tla.ovptl.uci.edu/api/graph/package/4090874>

Analysis

Curricular Analytics Report

Powered by the CurricularAnalytics Framework (Heileman, Abdallah, et al.)

Rule-Based Analysis

1. CURRICULAR ANALYTICS — METRIC DEFINITIONS

Based on the CurricularAnalytics framework — Heileman, Abdallah, et al.

A curriculum is modeled as a directed graph $G_c = (V, E)$, where each course is a vertex and each prerequisite or corequisite relationship is a directed edge.

- **Blocking Factor $b(v)$:** The number of courses reachable from course v — i.e., the courses that directly or transitively depend on v . A high blocking factor identifies a “gateway” course.
- **Delay Factor $d(v)$:** The number of vertices on the longest path passing through v . Represents the longest prerequisite chain involving v .
- **Structural Complexity $h(v) = d(v) + b(v)$:** Combines delay and blocking into a single measure. High complexity means the course is both hard to reach and critical to downstream progress.
- **Centrality $q(v)$:** Sum of the lengths of all long paths (source-to-sink) passing through v . A highly central course connects foundational prerequisites to advanced discipline-specific courses.

2. DEGREE PLAN OVERVIEW

Student ID	05208524	Entry	2020-21
Entry School	School of Physical Sciences	Entry Major	Applied Physics
Grad	2023-24	Grad Yrs	4.00
Curriculum	BS App Phy	Gender	M
Transferred	N	GPA	3.64
URM	N	FirstGen	Y
LowIncome	N	Avg Complexity	6.71
Avg Blocking	2.58	Avg Delay	4.13
Avg Centrality	17.81		

This degree plan spans **15** terms with **48** courses and **162** total credit hours.

Curriculum-level metrics — Total Complexity: **322.00**, Total Blocking: **124.00**, Total Delay: **198.00**.

Per-course averages — Complexity: **6.71**, Blocking: **2.58**, Delay: **4.13**, Centrality: **17.81**.

3. STUDENT PATH ANALYSIS

The student carried heavy course loads (≥ 16 credits) in **Term 3** (19 cr), **Term 4** (16 cr), **Term 5** (17 cr). Heavy terms increase the risk of course failure.

Lighter terms (≤ 12 credits): **Term 1** (4 cr), **Term 2** (4 cr), **Term 6** (4 cr), **Term 7** (4 cr), **Term 13** (12 cr), **Term 14** (1 cr), **Term 15** (10 cr).

3.1 BLOCKING FACTOR ANALYSIS

The blocking factor $b(v)$ counts how many courses are reachable from v . A high blocking factor identifies “gateway” courses — failing or delaying such a course cascades into delays for every dependent course.

Peak blocking occurred in **Term 3** (aggregate: 42.00). Curriculum total blocking factor: **124.00**.

3. STUDENT PATH ANALYSIS

The student carried heavy course loads (≥ 16 credits) in **Term 3** (19 cr), **Term 4** (16 cr), **Term 5** (17 cr). Heavy terms increase the risk of course failure.

Lighter terms (≤ 12 credits): **Term 1** (4 cr), **Term 2** (4 cr), **Term 6** (4 cr), **Term 7** (4 cr), **Term 13** (12 cr), **Term 14** (1 cr), **Term 15** (10 cr).

3.1 BLOCKING FACTOR ANALYSIS

The blocking factor $b(v)$ counts how many courses are reachable from v . A high blocking factor identifies “gateway” courses — failing or delaying such a course cascades into delays for every dependent course.

Peak blocking occurred in **Term 3** (aggregate: 42.00). Curriculum total blocking factor: **124.00**.

Key gateway/bottleneck courses (blocking > 3.9):

- **PHYSICS 7C** — $b(v) = 25.00$: Blocks 25 downstream courses.
- **PHYSICS 7D** — $b(v) = 13.00$: Blocks 13 downstream courses.
- **MATH 2D** — $b(v) = 12.00$: Blocks 12 downstream courses.
- **MATH 2A** — $b(v) = 9.00$: Blocks 9 downstream courses.
- **PHYSICS 7E** — $b(v) = 8.00$: Blocks 8 downstream courses.

3.2 DELAY FACTOR ANALYSIS

The delay factor $d(v)$ is the length of the longest path passing through v — the critical path. Any failure on this path extends time-to-degree by at least one term.

Peak delay occurred in **Term 11** (aggregate: 28.00). Curriculum total delay factor: **198.00**.

Critical-path courses (delay > 6.2):

- **MATH 2A** — $d(v) = 8.00$: Prerequisite chain of 8 courses.
- **PHYSICS 7C** — $d(v) = 8.00$: Prerequisite chain of 8 courses.
- **PHYSICS 7D** — $d(v) = 8.00$: Prerequisite chain of 8 courses.
- **EECS 10** — $d(v) = 8.00$: Prerequisite chain of 8 courses.
- **EECS 70A** — $d(v) = 8.00$: Prerequisite chain of 8 courses.

3.3 STRUCTURAL COMPLEXITY ANALYSIS

Structural complexity $h(v) = d(v) + b(v)$ combines delay and blocking into a single measure. High complexity means the course is both hard to reach and a gateway to many downstream courses.

Peak complexity occurred in **Term 3** (aggregate: 63.00). Curriculum total complexity: **322.00**.

Most complex courses (complexity > 10.1):

- **PHYSICS 7C** — $h(v) = 33.00$ (blocking: 25.00 + delay: 8.00). Major structural chokepoint.
- **PHYSICS 7D** — $h(v) = 21.00$ (blocking: 13.00 + delay: 8.00). Major structural chokepoint.
- **MATH 2A** — $h(v) = 17.00$ (blocking: 9.00 + delay: 8.00). Major structural chokepoint.
- **MATH 2D** — $h(v) = 16.00$ (blocking: 12.00 + delay: 4.00). Major structural chokepoint.
- **EECS 10** — $h(v) = 16.00$ (blocking: 8.00 + delay: 8.00). Major structural chokepoint.

3.4 CENTRALITY ANALYSIS

Centrality $q(v)$ measures how central a course is within the prerequisite network — computed as the sum of all long-path lengths passing through v .

Most central courses (centrality > 26.7):

- **EECS 70B** — $q(v) = 145.00$: Critical junction linking foundational to advanced coursework.
- **EECS 70A** — $q(v) = 110.00$: Critical junction linking foundational to advanced coursework.
- **EECS 10** — $q(v) = 104.00$: Critical junction linking foundational to advanced coursework.

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- **EECS 70A** — $q(v) = 110.00$: Critical junction linking foundational to advanced coursework.
- **EECS 10** — $q(v) = 104.00$: Critical junction linking foundational to advanced coursework.
- **PHYSICS 7D** — $q(v) = 101.00$: Critical junction linking foundational to advanced coursework.
- **EECS 170A** — $q(v) = 100.00$: Critical junction linking foundational to advanced coursework.

4. RECOMMENDATIONS

- ▶ Gateway courses (**PHYSICS 7C, PHYSICS 7D, MATH 2D**) should be prioritized early. Their high blocking factors mean failure cascades into many downstream delays.
- ▶ Terms **3, 4, 5** combine heavy credit loads with high-complexity courses — consider redistributing to reduce risk.
- ▶ Critical-path courses (**MATH 2A, PHYSICS 7C, PHYSICS 7D**) sit on the longest prerequisite chains and should never be deferred.
- ▶ Highly central courses (**EECS 70B, EECS 70A, EECS 10**) are critical junctions. Academic support should focus on these courses.